Intensity Transformations and Spatial Filtering

Histogram

image enhancement techniques working in the spatial domain:

- What is image enhancement?
- Different kinds of image enhancement
- Histogram processing
- Point processing
- Neighbourhood operations

What Is Image Enhancement?

Image enhancement is the process of making images more useful

The reasons for doing this include:

- Highlighting interesting detail in images
- Removing noise from images
- Making images more visually appealing

Image Enhancement Examples

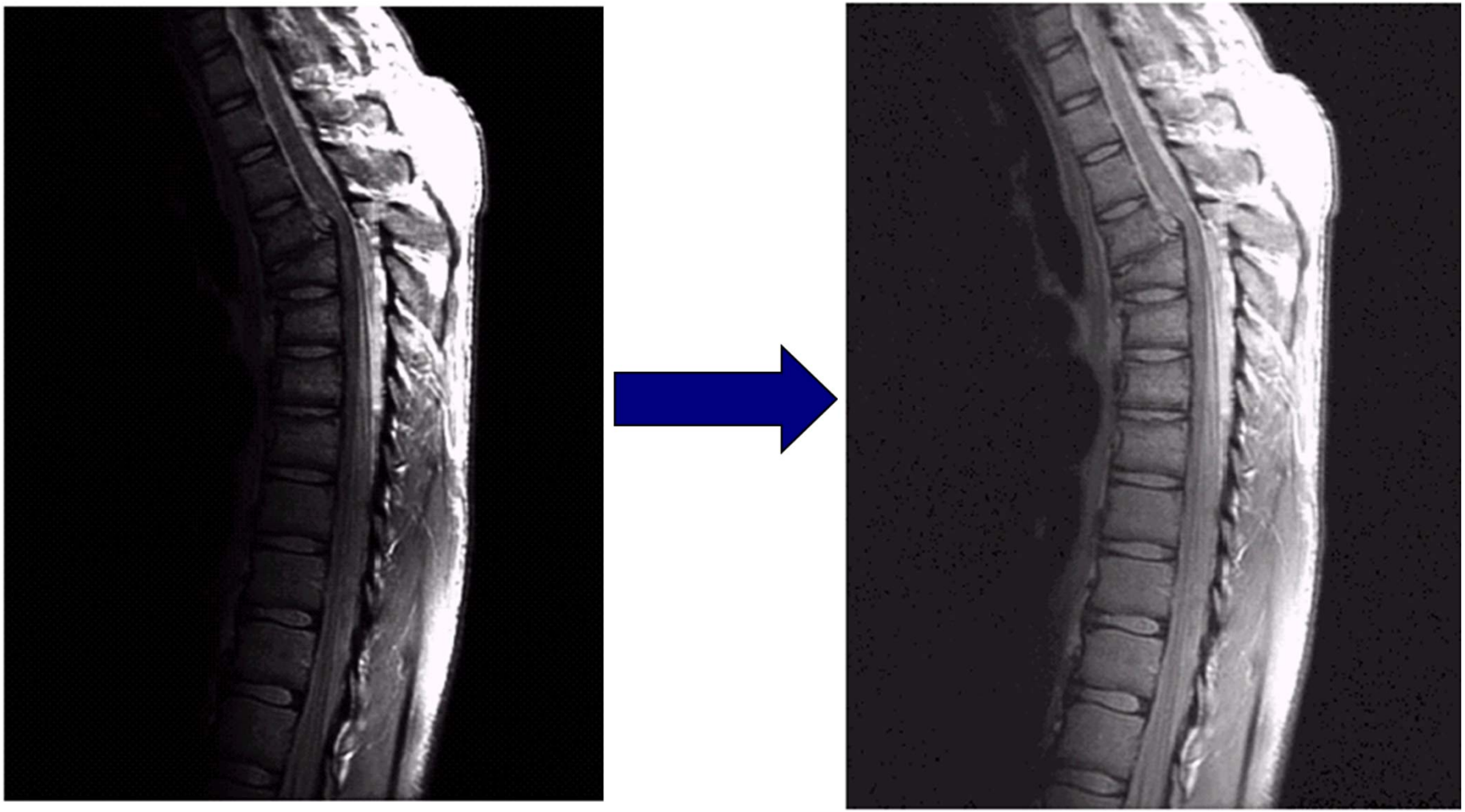
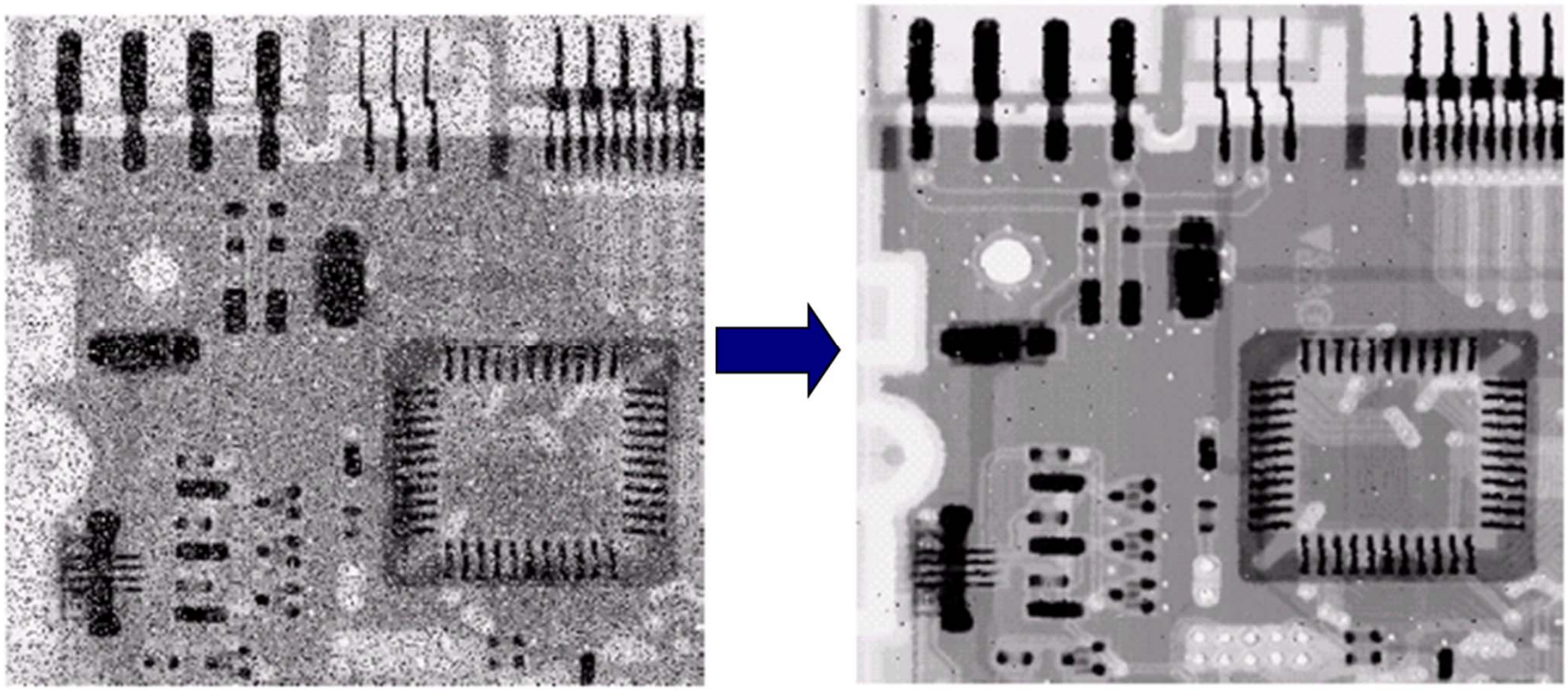


Image Enhancement Examples



Image Enhancement Examples



Spatial & Frequency Domains

There are two broad categories of image enhancement techniques

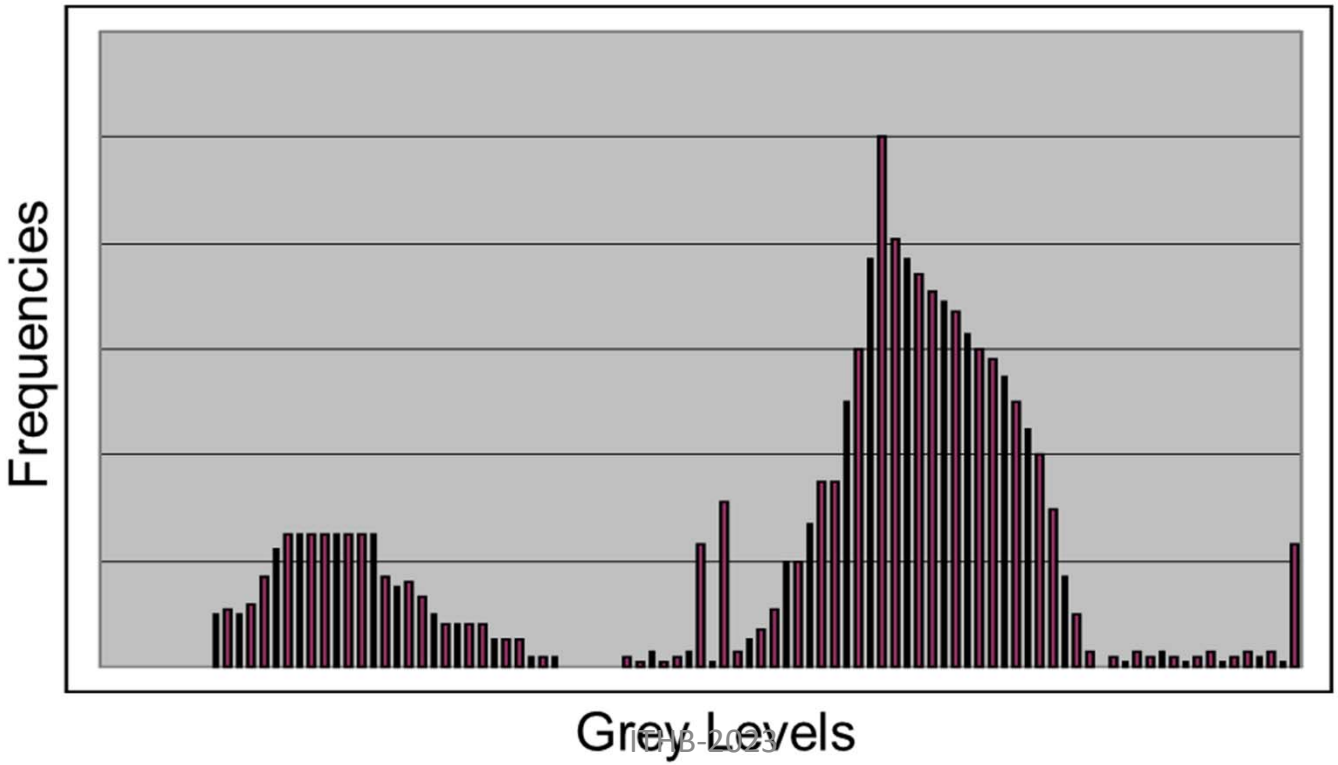
- Spatial domain techniques
  - Direct manipulation of image pixels
- Frequency domain techniques
  - Manipulation of Fourier transform or wavelet transform of an image

For the moment we will concentrate on techniques that operate in the spatial domain

Image Histograms

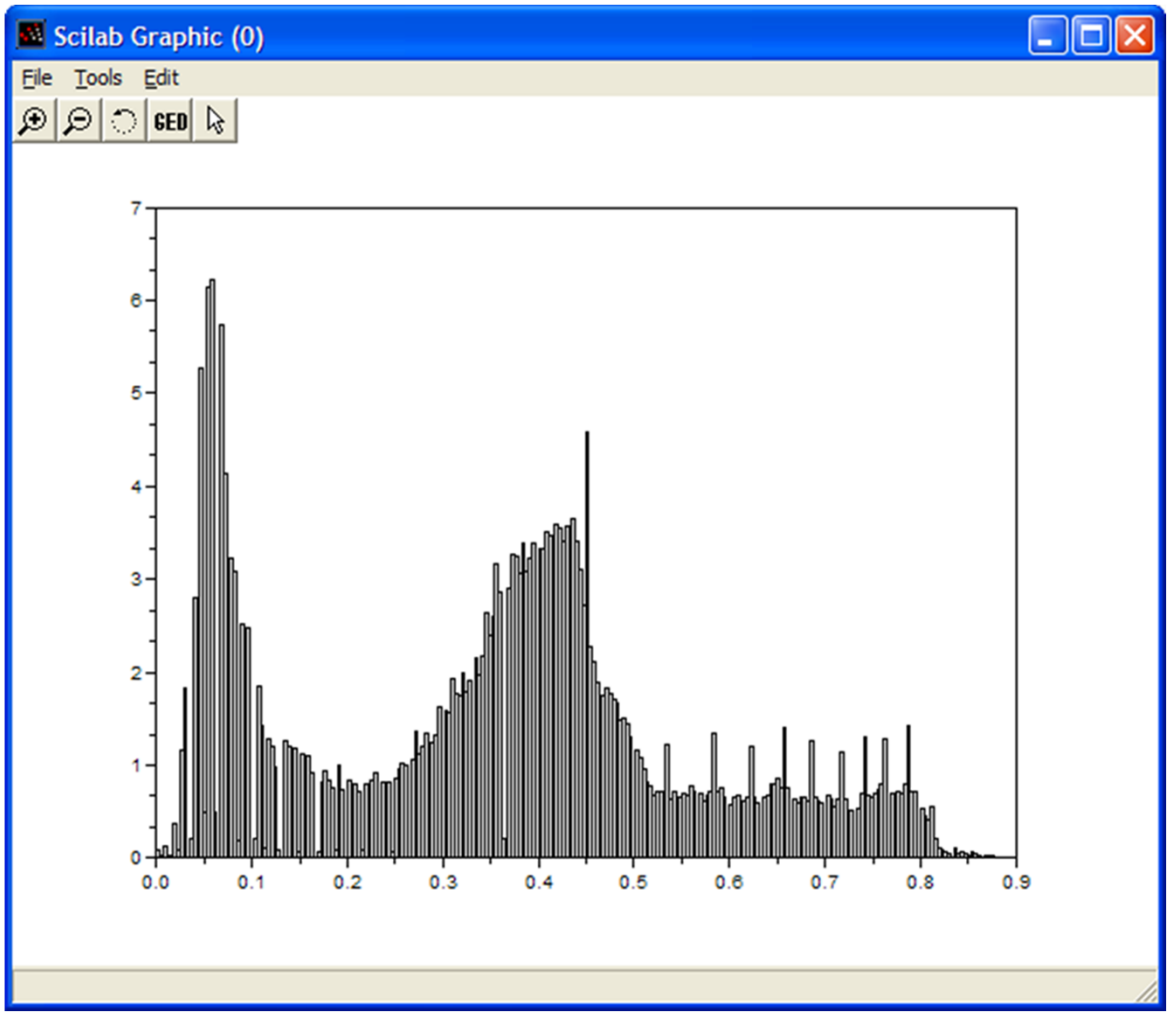
The histogram of an image shows us the distribution of grey levels in the image

Massively useful in image processing, especially in segmentation

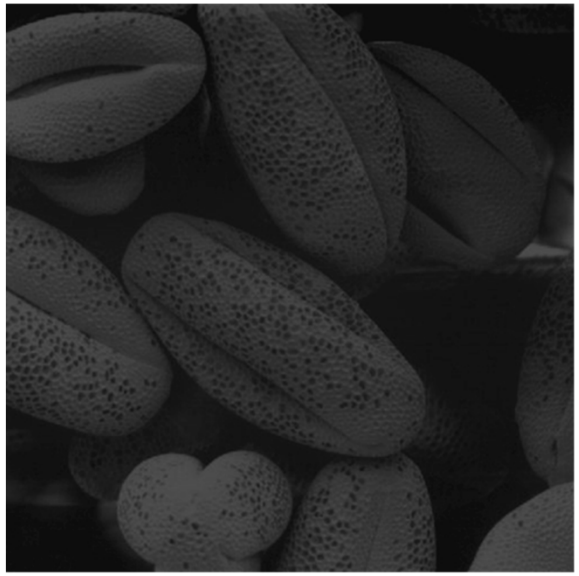


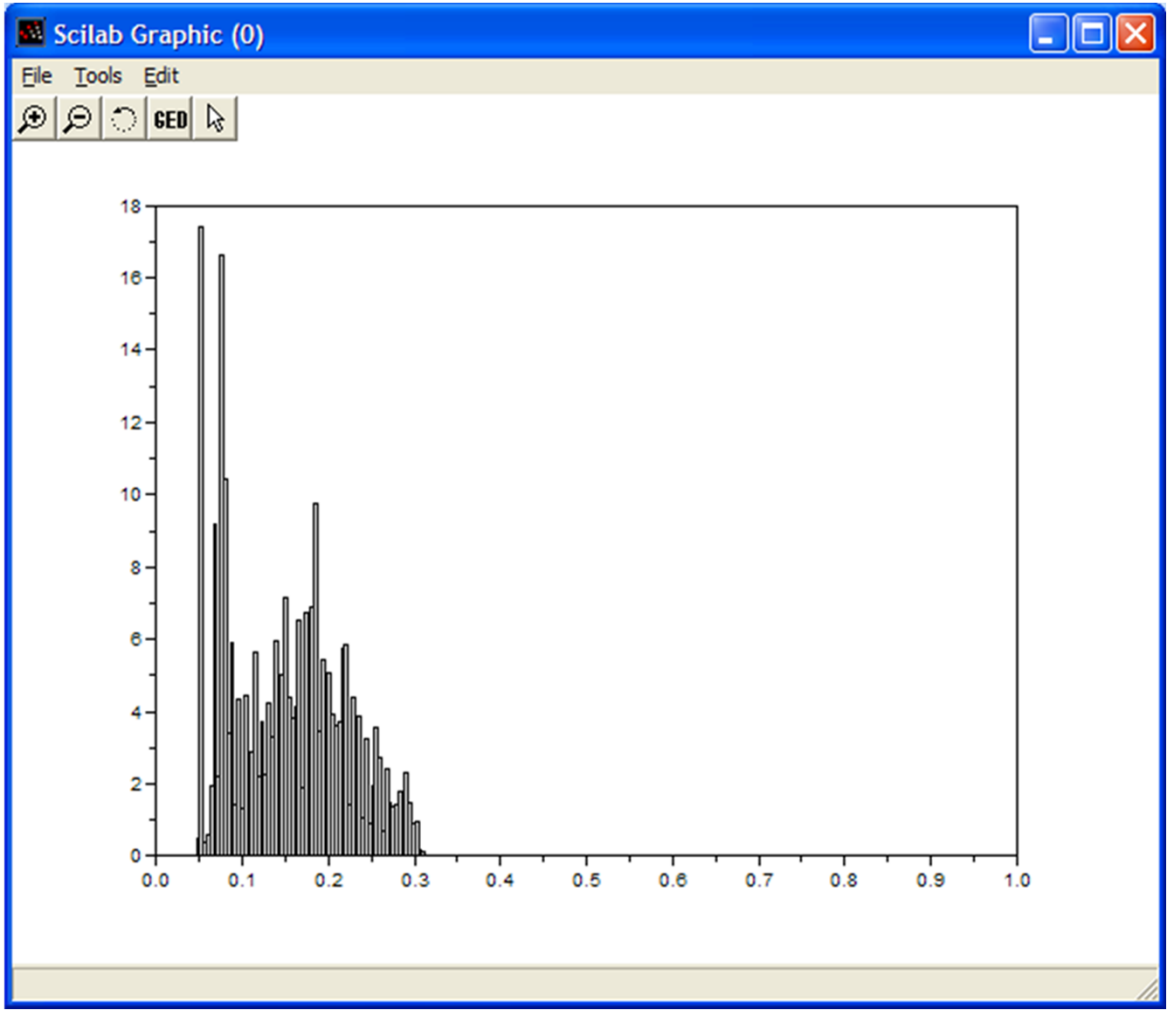
Histograms Example





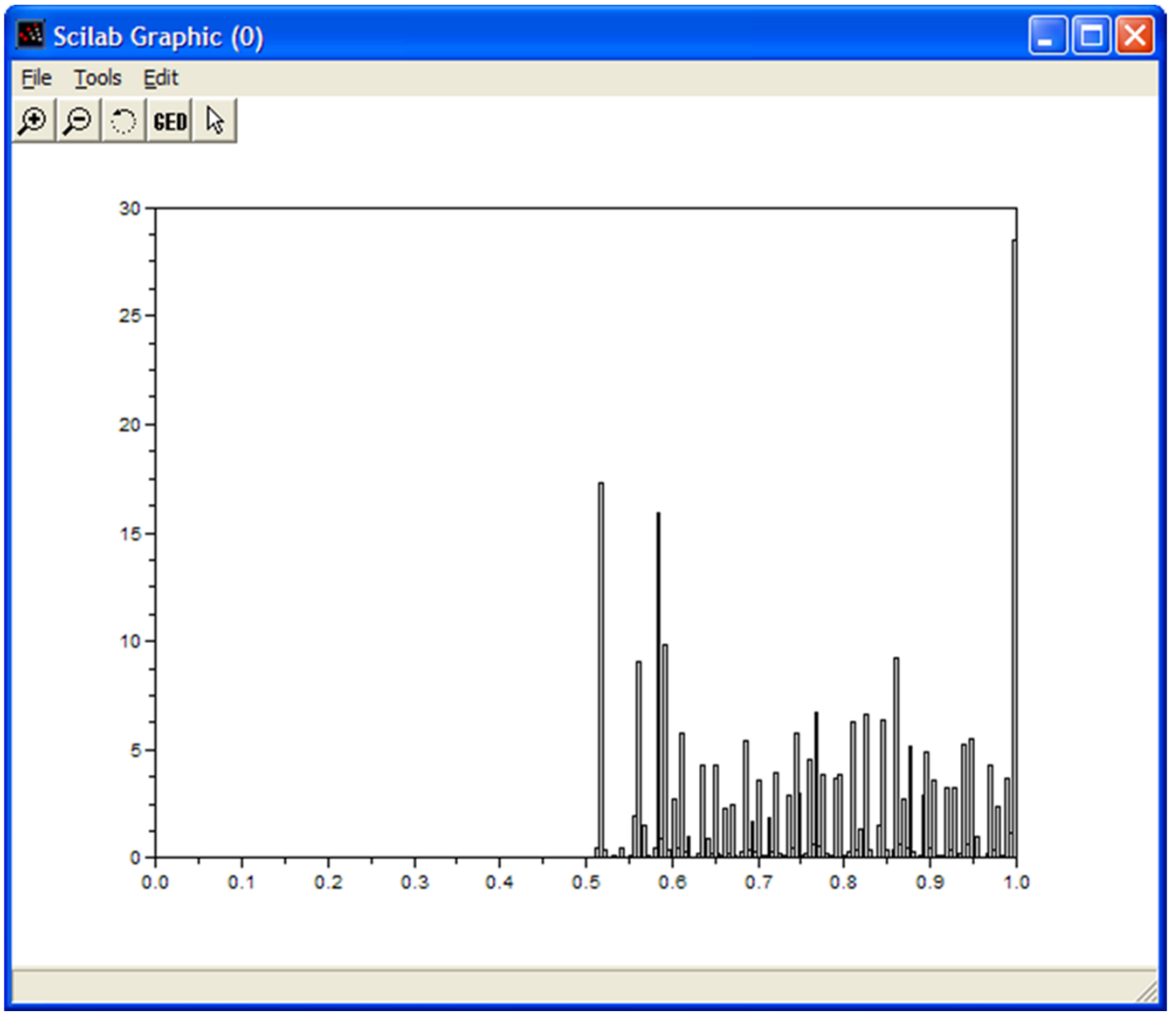
Histograms Example



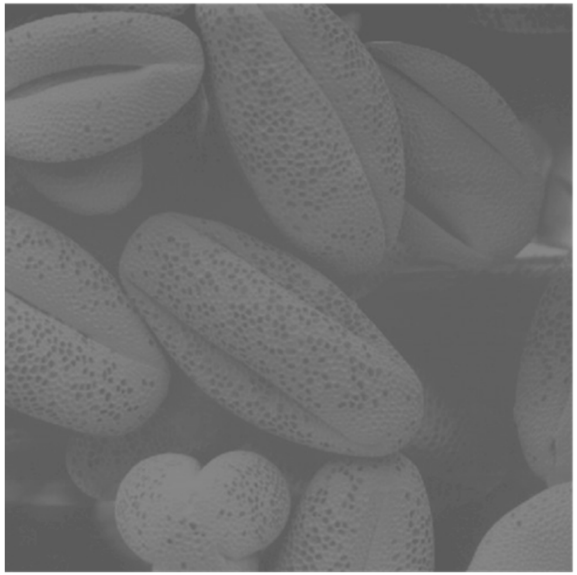


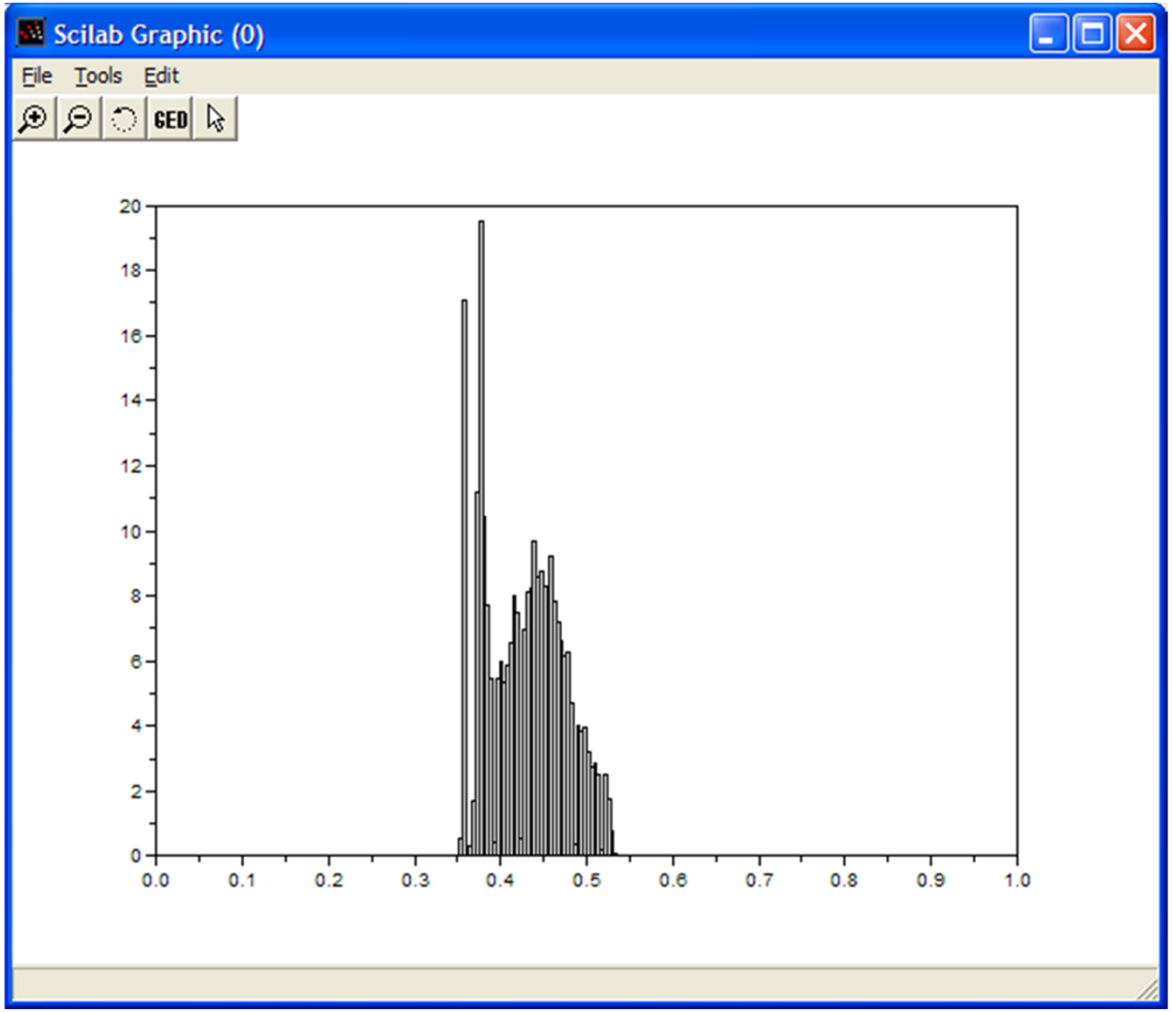
Histograms Example





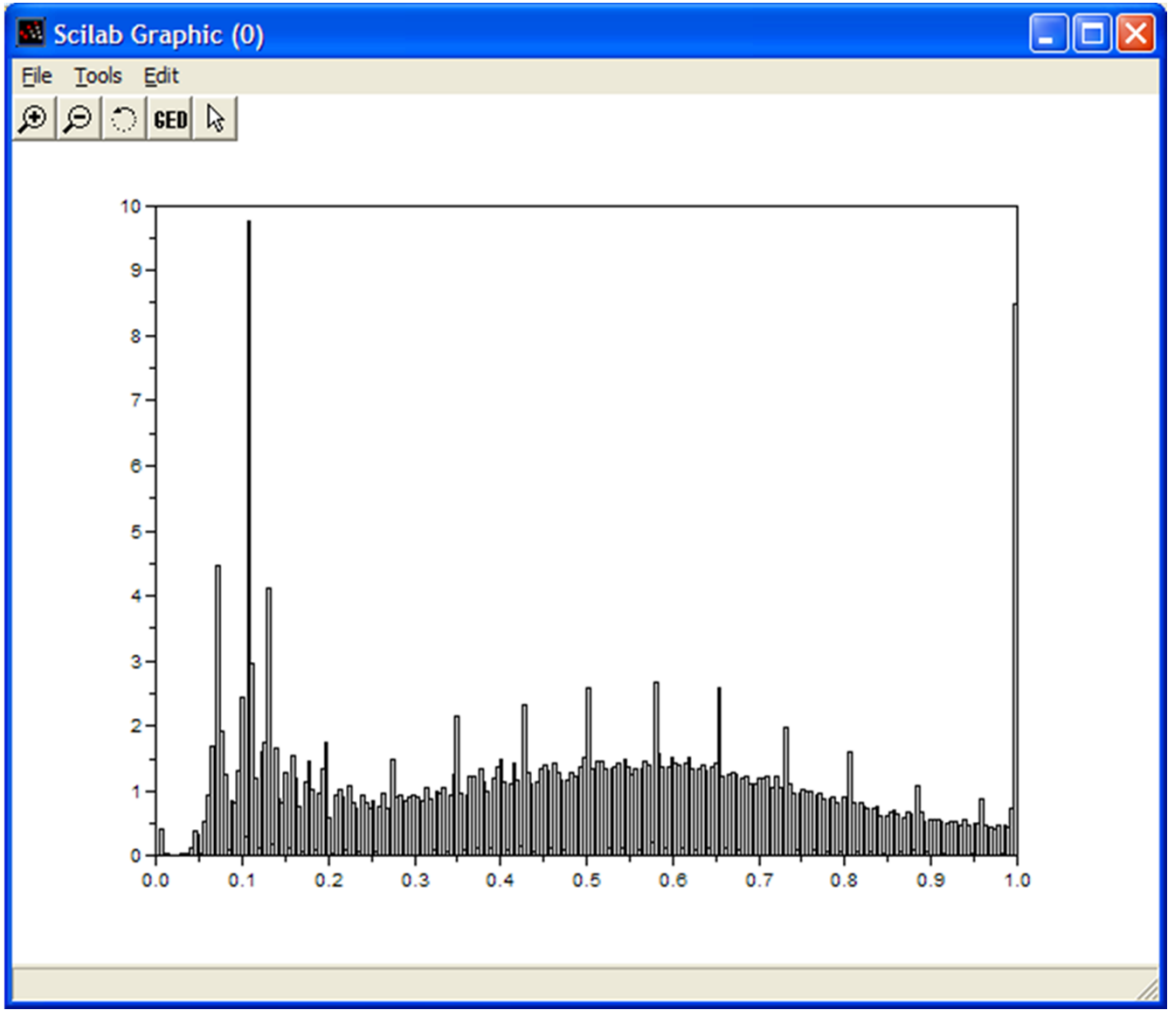
Histograms Example





Histograms Example





Contrast Stretching

- Low-contrast images can result from poor illumination, lack of dynamic range in the imaging sensor, or even the wrong setting of a lens aperture during image acquisition.
- Contrast stretching expands the range of intensity levels in an image so that it spans the ideal full intensity range of the recording medium or display device

Contrast Stretching

a b
c d

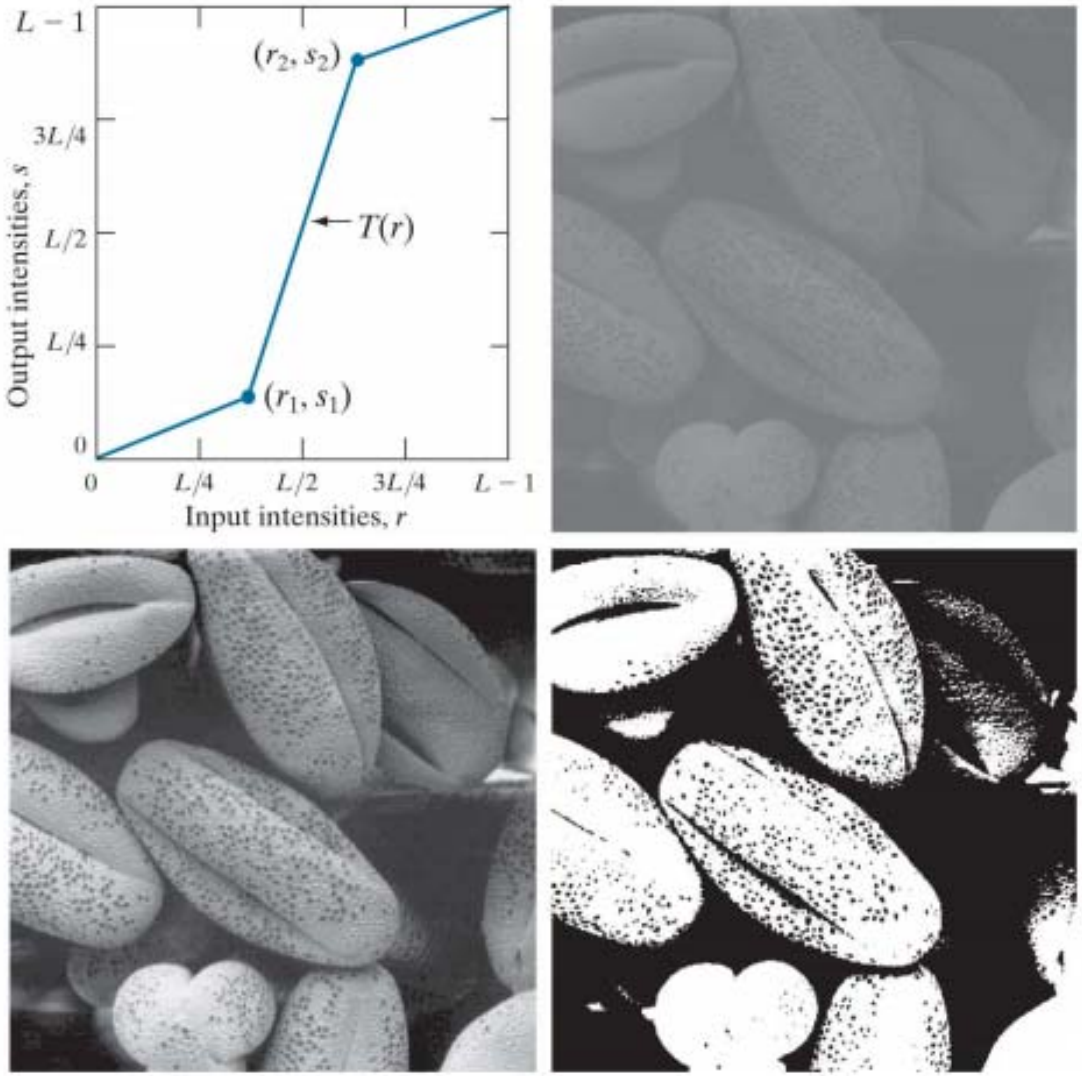


FIGURE 3.10
Contrast stretching. (a) Piecewise linear transformation function. (b) A low-contrast electron microscope image of pollen, magnified 700 times. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

Linear Stretching

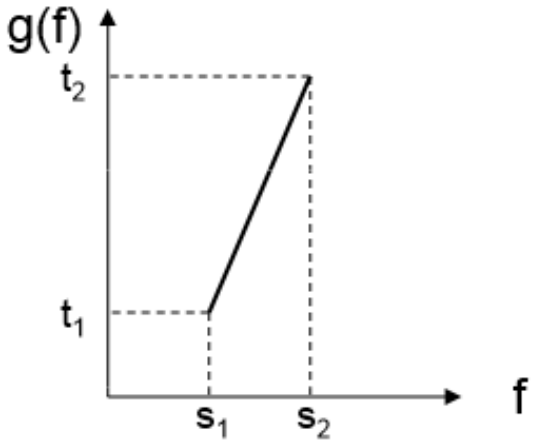
Enhance the dynamic range by linearly stretching the original gray levels to the range of

$$g(f) = af + b$$

$$[s_1 \quad s_2] \Rightarrow [t_1 \quad t_2]$$

$$g(f) = (f - s_1) * (t_2 - t_1) / (s_2 - s_1) + t_1$$

$$a = (t_2 - t_1) / (s_2 - s_1), \quad b = t_1 - s_1 * (t_2 - t_1) / (s_2 - s_1)$$



Example

- The original gray levels are [100, 150].
- The target gray levels are [0, 255].
- $g(f) = ((f - 100) / 50) * 255, \quad for\ 100 \le f \le 150.$

Histograms

- histogram of f is defined as

$$h(r_k) = n_k \quad \text{for } k = 0, 1, 2, ..., L-1$$

- where is the number of pixels in f with intensity and the subdivisions of the intensity scale are called histogram bins. Similarly, the normalized histogram of f is defined as

$$p(r_k) = \frac{h(r_k)}{MN} = \frac{n_k}{MN}$$

Histogram Equalization

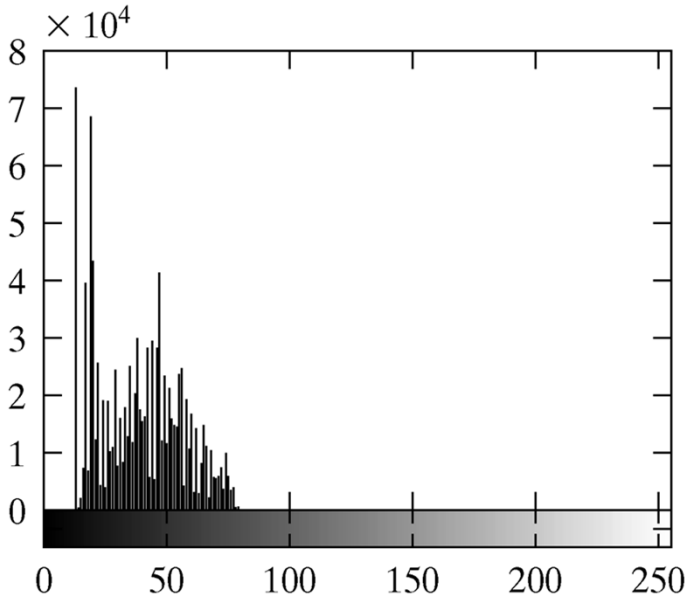
Spreading out the frequencies in an image (or equalising the image) is a simple way to improve dark or washed out images

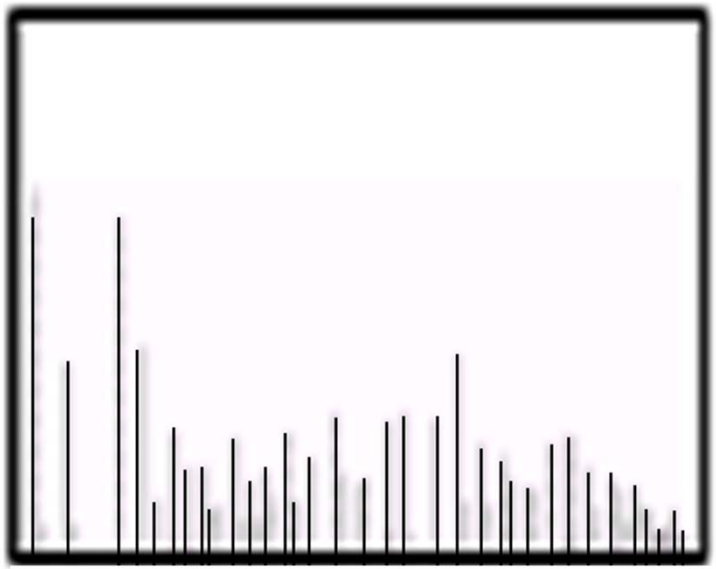
The formula for histogram equalisation is given where

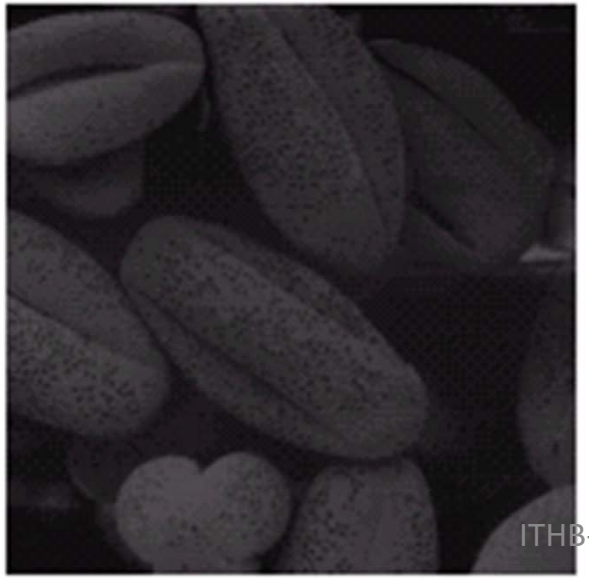
- r_k: input intensity
- s_k: processed intensity
- k: the intensity range (e.g 0.0 – 1.0)
- n_j: the frequency of intensity j
- n: the sum of all frequencies

$$
\begin{aligned}
s_k &= T(r_k) \\
&= \sum_{j=1}^{k} p_r(r_j) \\
&= \sum_{j=1}^{k} \frac{n_j}{n}
\end{aligned}
$$

Histogram Equalization



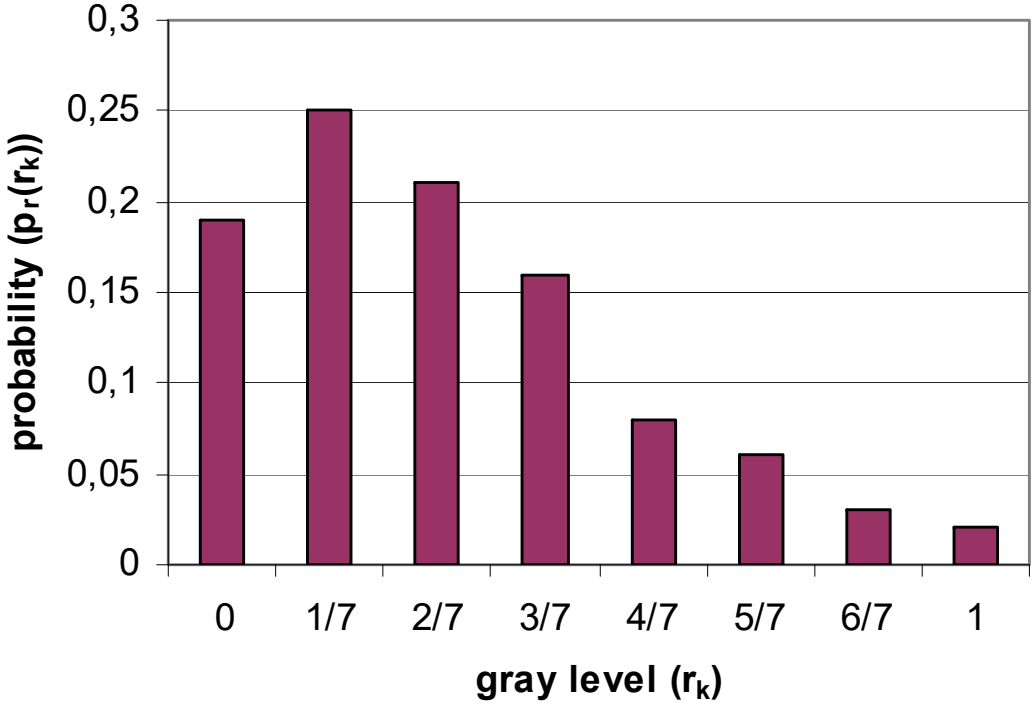






Example

r_k	n_k	$p_r(r_k)=n_k/n$
$r_0=0$	790	0,19
$r_1=1/7$	1023	0,25
$r_2=2/7$	850	0,21
$r_3=3/7$	656	0,16
$r_4=4/7$	329	0,08
$r_5=5/7$	245	0,06
$r_6=6/7$	122	0,03
$r_7=1$	81	0,02



Transformation Function

$$s_0 = T(r_0) = \sum_{j=0}^{0} p_r(r_j) = p_r(r_0) = 0.19$$

$$s_1 = T(r_1) = \sum_{j=0}^{1} p_r(r_j) = p_r(r_0) + p_r(r_1) = 0.44$$

$$s_2 = T(r_2) = \sum_{j=0}^{2} p_r(r_j) = p_r(r_0) + p_r(r_1) + p_r(r_2) = 0.65$$

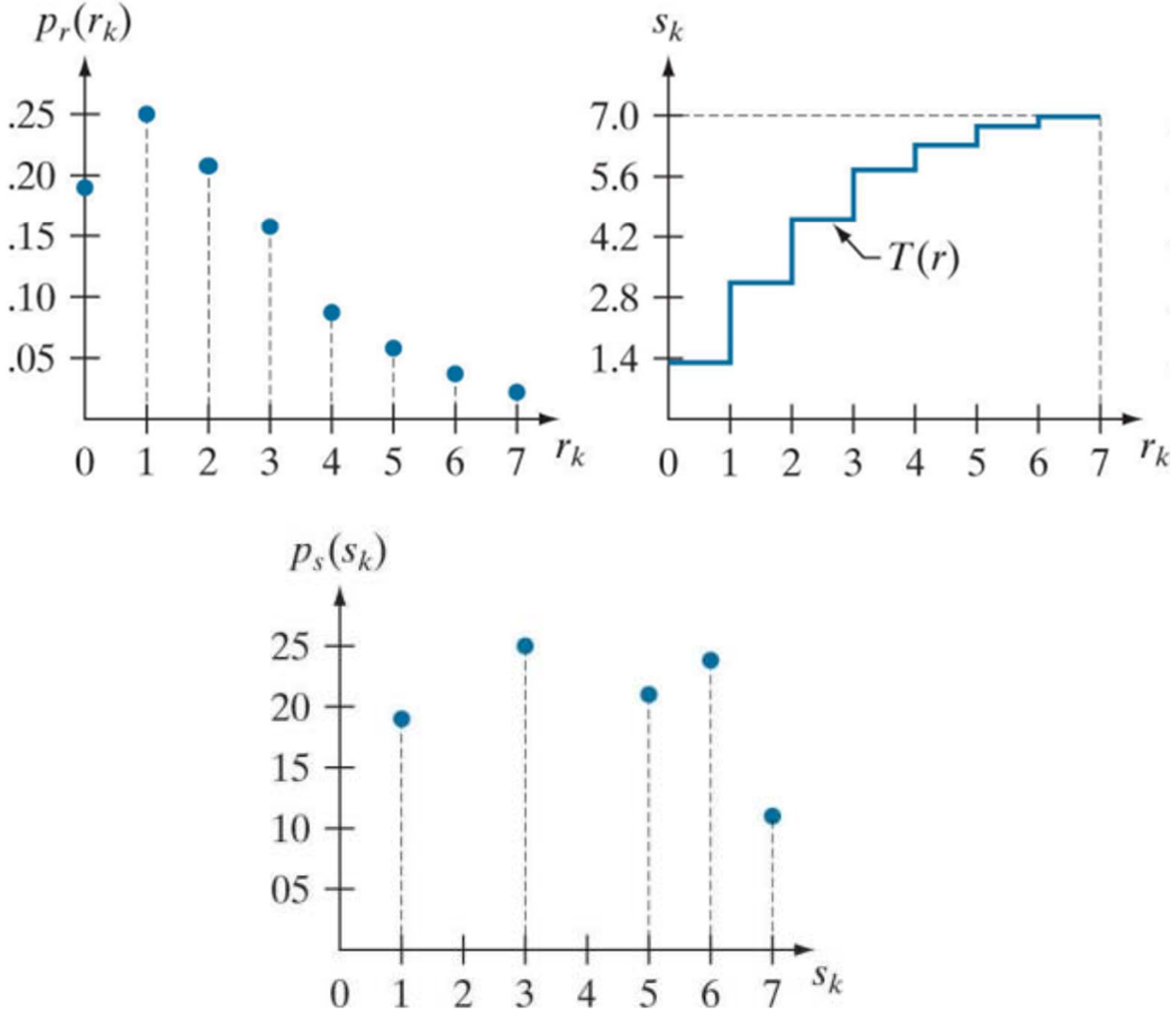
$$s_3 = T(r_3) = \sum_{j=0}^{3} p_r(r_j) = 0.81; \qquad s_4 = T(r_4) = \sum_{j=0}^{4} p_r(r_j) = 0.89$$

$$s_5 = T(r_5) = \sum_{j=0}^{5} p_r(r_j) = 0.95; \qquad s_6 = T(r_6) = \sum_{j=0}^{6} p_r(r_j) = 0.98$$

$$s_7 = T(r_7) = \sum_{j=0}^{7} p_r(r_j) = 1.00$$

$$s_0 = T(r_0) = 7 \sum_{j=0}^{0} p_r(r_j) = 7p_r(r_0) = 1.33$$

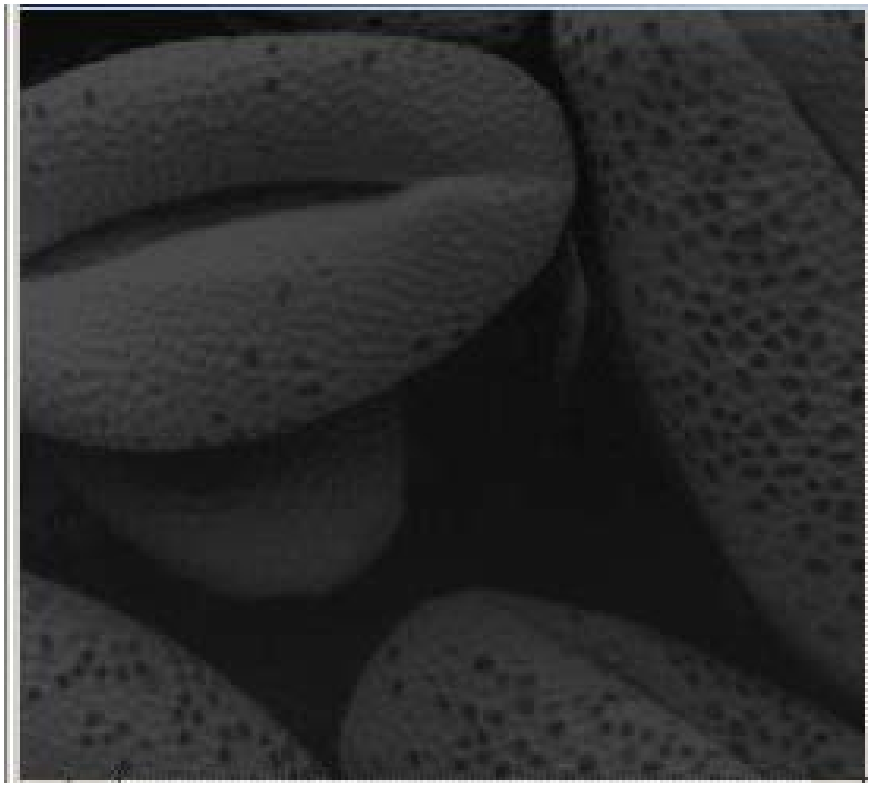
r_k	n_k	$p_r(r_k)=n_k/n$	S_k	S_k x 7	Normal(S_k)
$r_0=0$	790	0,19	0,19	1,33 ≅ 1	$s_0=1/7$
$r_1=1/7$	1023	0,25	0,44	3,08 ≅ 3	$s_1=3/7$
$r_2=2/7$	850	0,21	0,65	4,55 ≅ 5	$s_2=5/7$
$r_3=3/7$	656	0,16	0,81	5,67 ≅ 6	$s_3=6/7$
$r_4=4/7$	329	0,08	0,89	6,23 ≅ 6	$s_4=6/7$
$r_5=5/7$	245	0,06	0,95	6,65 ≅ 7	$s_5=7/7$
$r_6=6/7$	122	0,03	0,98	6,86 ≅ 7	$s_6=7/7$
$r_7=1$	81	0,02	1,00	7	$s_7=1$

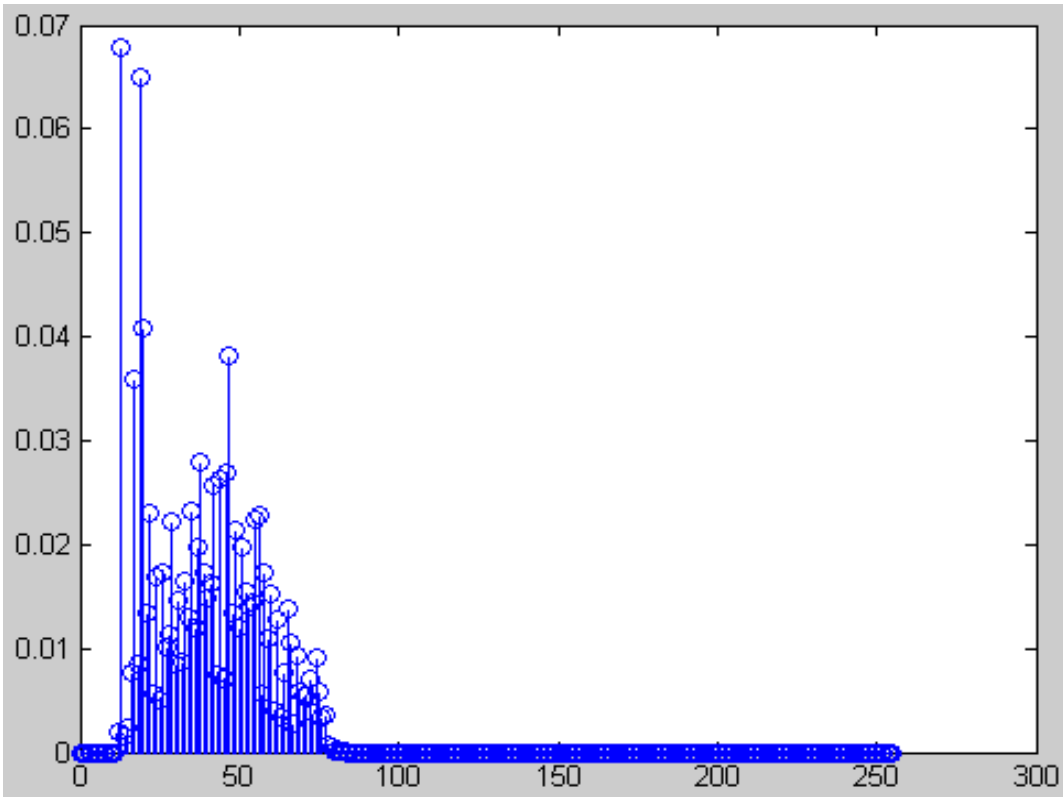


Histogram equalization. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Example

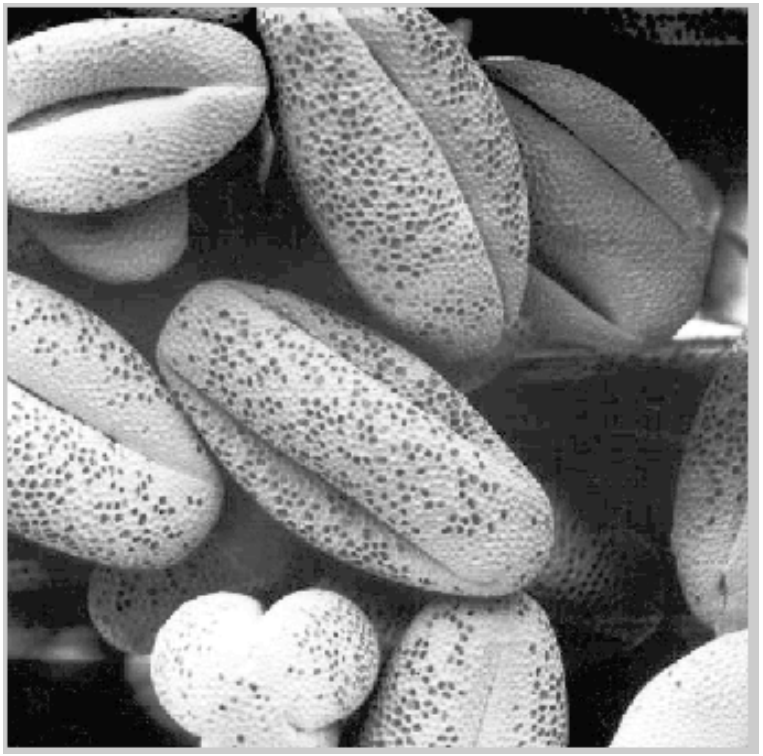
Original image and its histogram

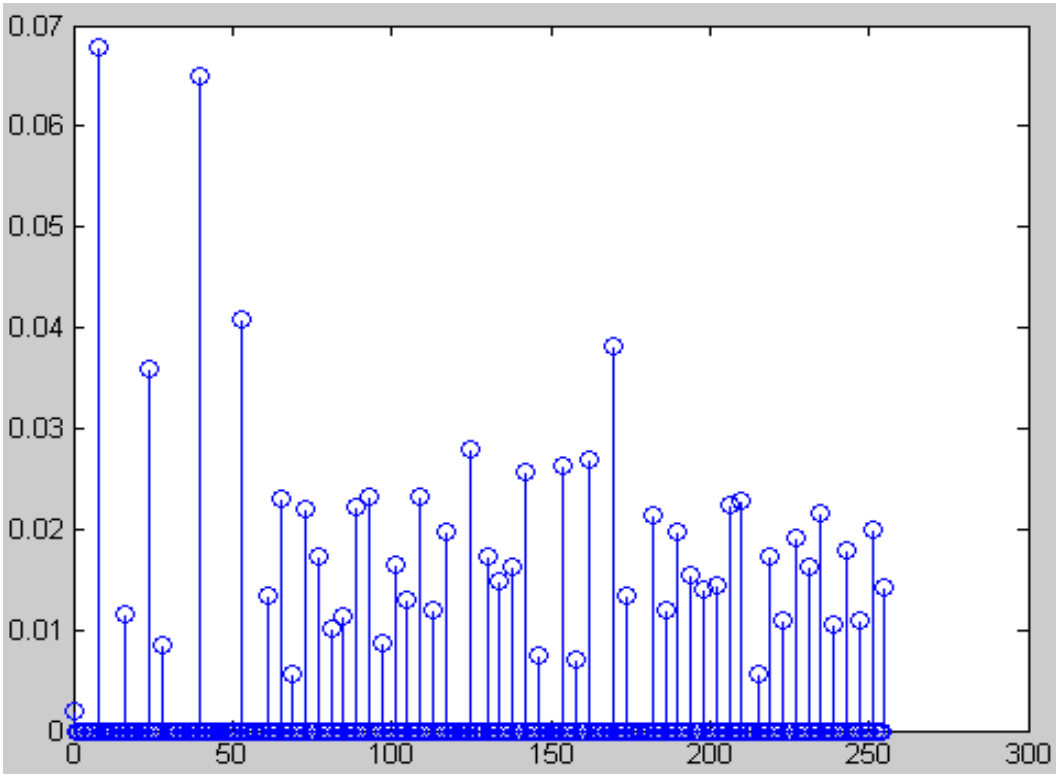




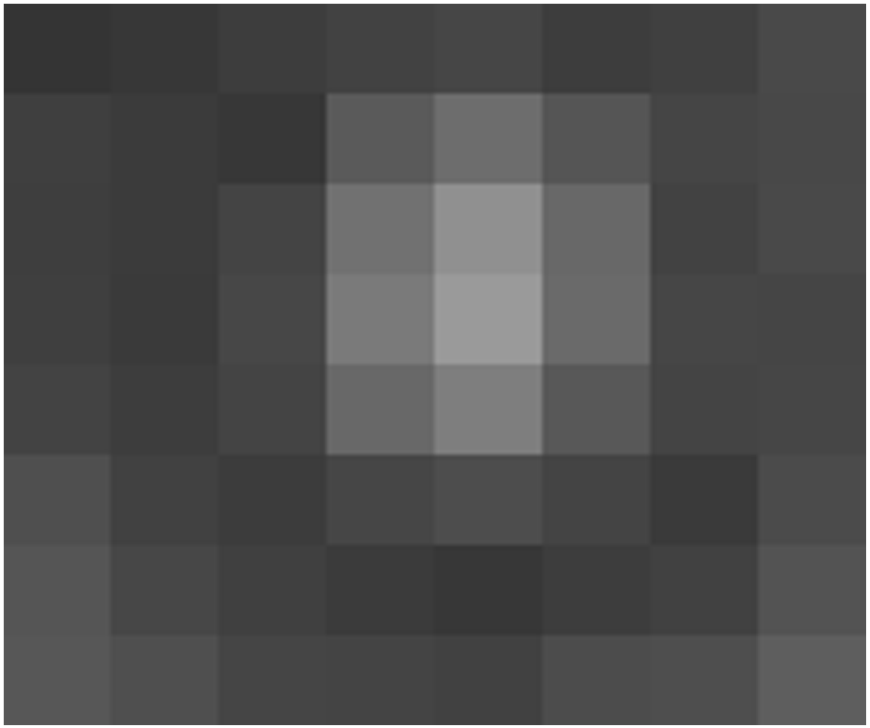

```
>> clear;
[ix,map]=imread('Fig3_15a.jpg');
imshow(ix)
figure;
ix=double(ix);
h=histogram(ix);
stem(0:255,h);
```

Histogram equalized image and its histogram

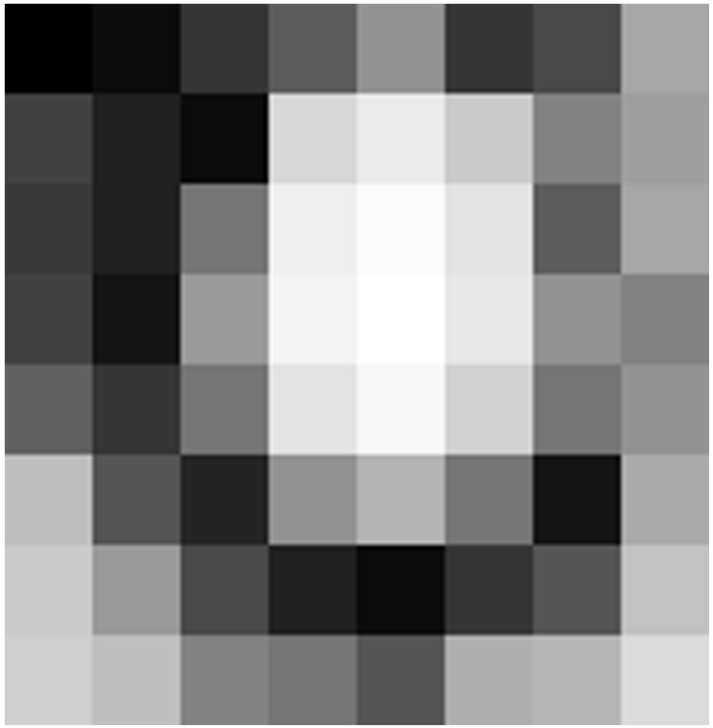





```
>> clear
>> [ix,map]=imread('Fig3_15a.jpg');
>> imshow(ix);
>> iy=histeq(ix);
>> figure
>> imshow(iy);
>> iy=double(iy);
>> hy=histogram(iy);
>> figure
>> stem(0:255,hy);
```

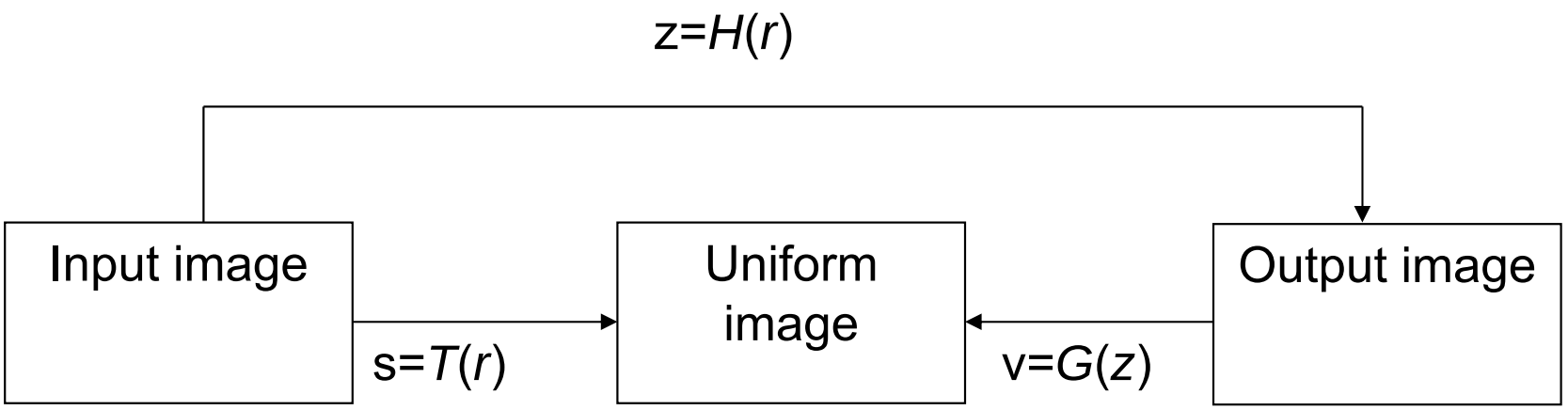


Original



Equalized

Histogram Specification

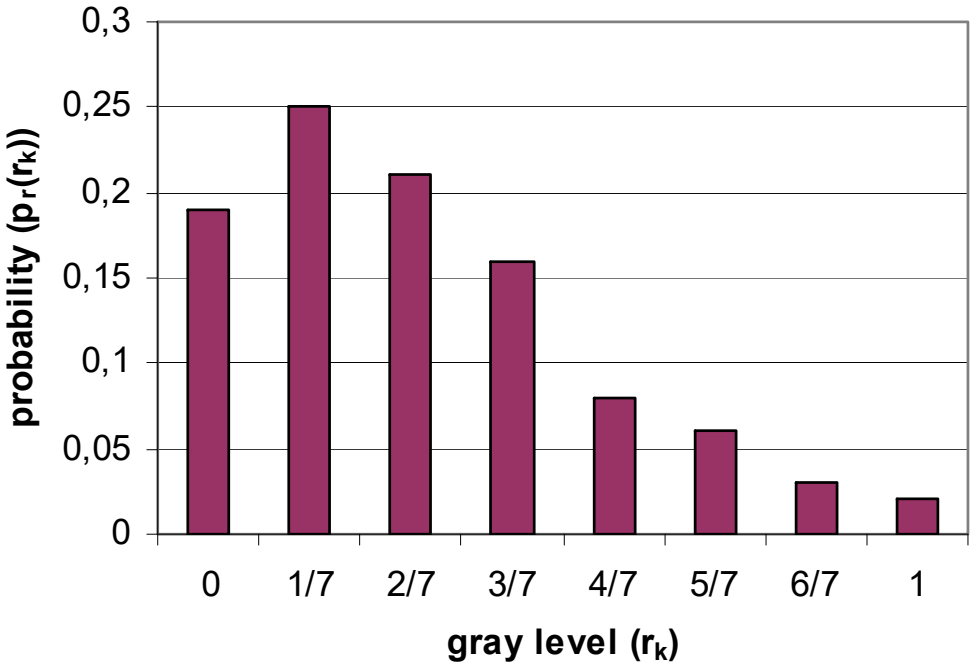


Example

Image 64x64 pixel, 8 color depth:

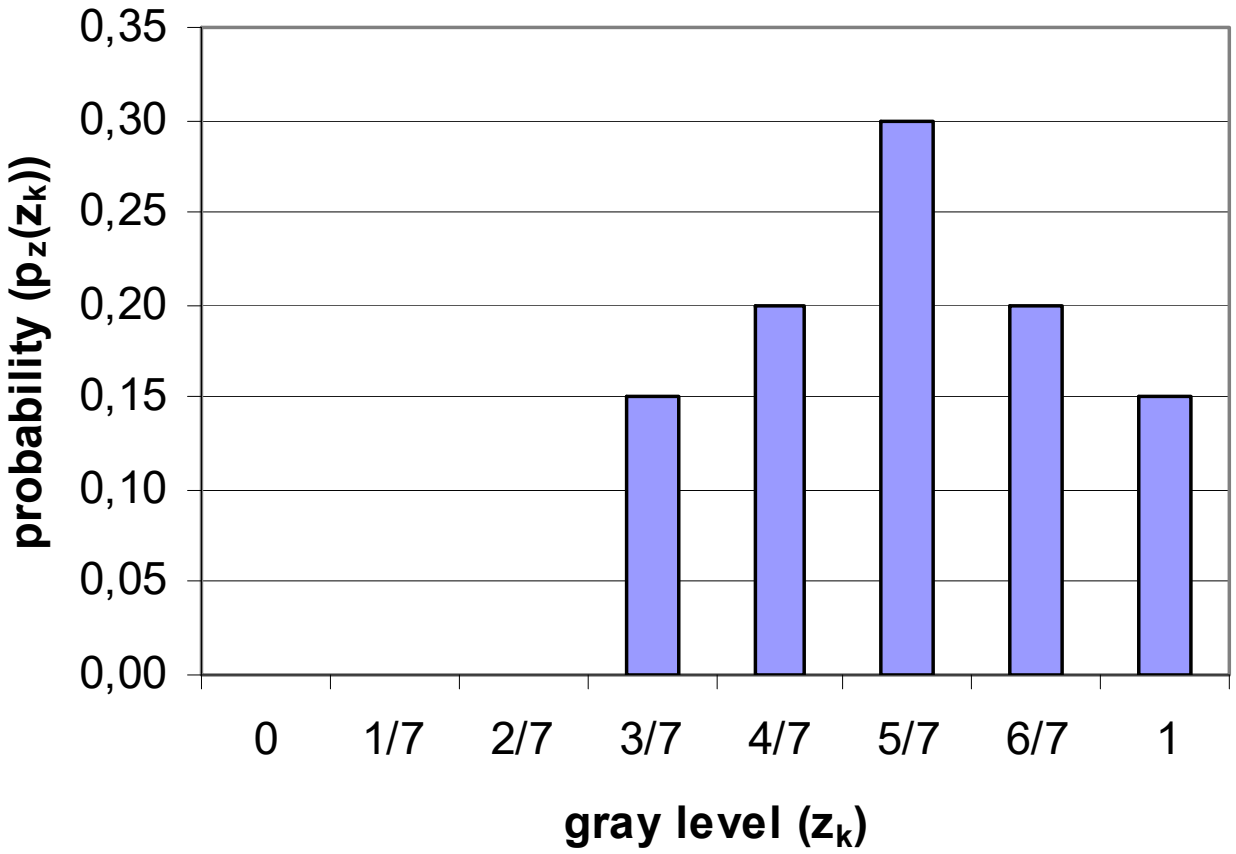
r_k	n_k	$p_r(r_k)=n_k/n$
$r_0=0$	790	0,19
$r_1=1/7$	1023	0,25
$r_2=2/7$	850	0,21
$r_3=3/7$	656	0,16
$r_4=4/7$	329	0,08
$r_5=5/7$	245	0,06
$r_6=6/7$	122	0,03
$r_7=1$	81	0,02

Histogram citra:



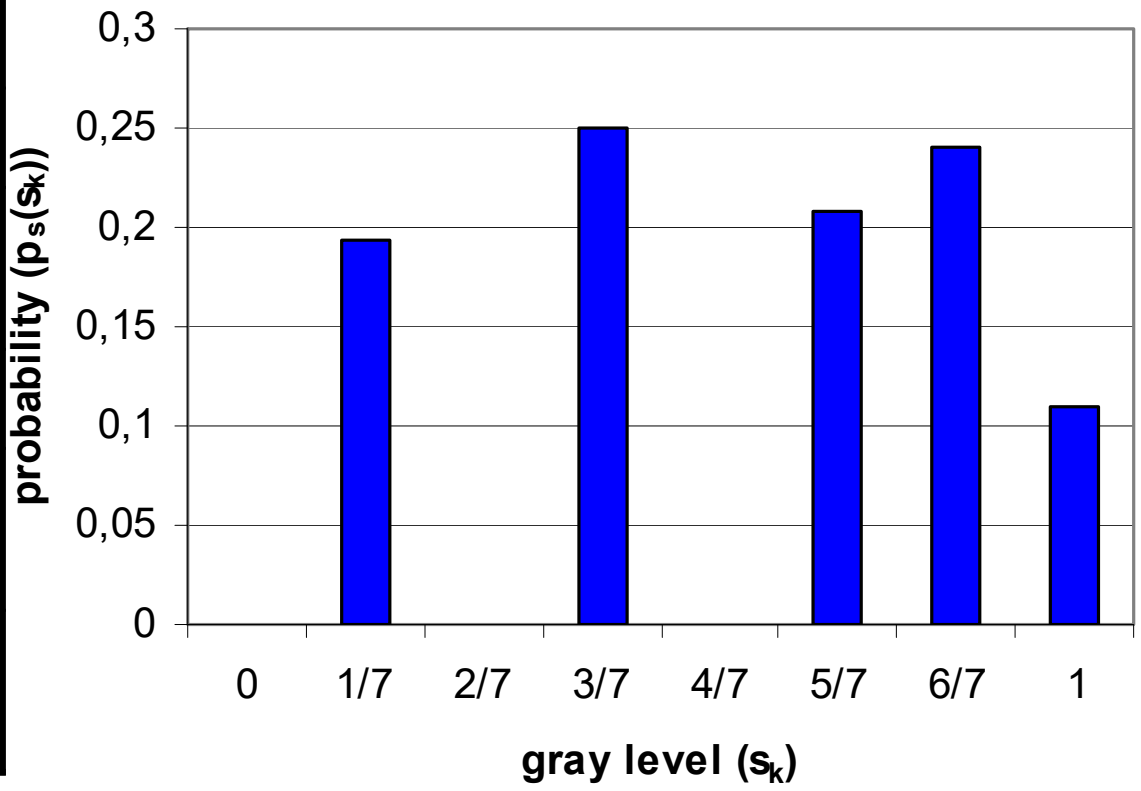
Histogram Spesification

z_k	$p_z(z_k)$
$z_0=0$	0,00
$z_1=1/7$	0,00
$z_2=2/7$	0,00
$z_3=3/7$	0,15
$z_4=4/7$	0,20
$z_5=5/7$	0,30
$z_6=6/7$	0,20
$z_7=1$	0,15



Step 1:
Equalisasi Histogram

$r_j \rightarrow s_k$	n_k	$p_s(s_k)$
$r_0 \rightarrow s_0 = 1/7$	790	0,19
$r_1 \rightarrow s_1 = 3/7$	1023	0,25
$r_2 \rightarrow s_2 = 5/7$	850	0,21
$r_3, r_4 \rightarrow s_3 = 6/7$	985	0,24
$r_5, r_6, r_7 \rightarrow s_4 = 7/7$	448	0,11



Step 2

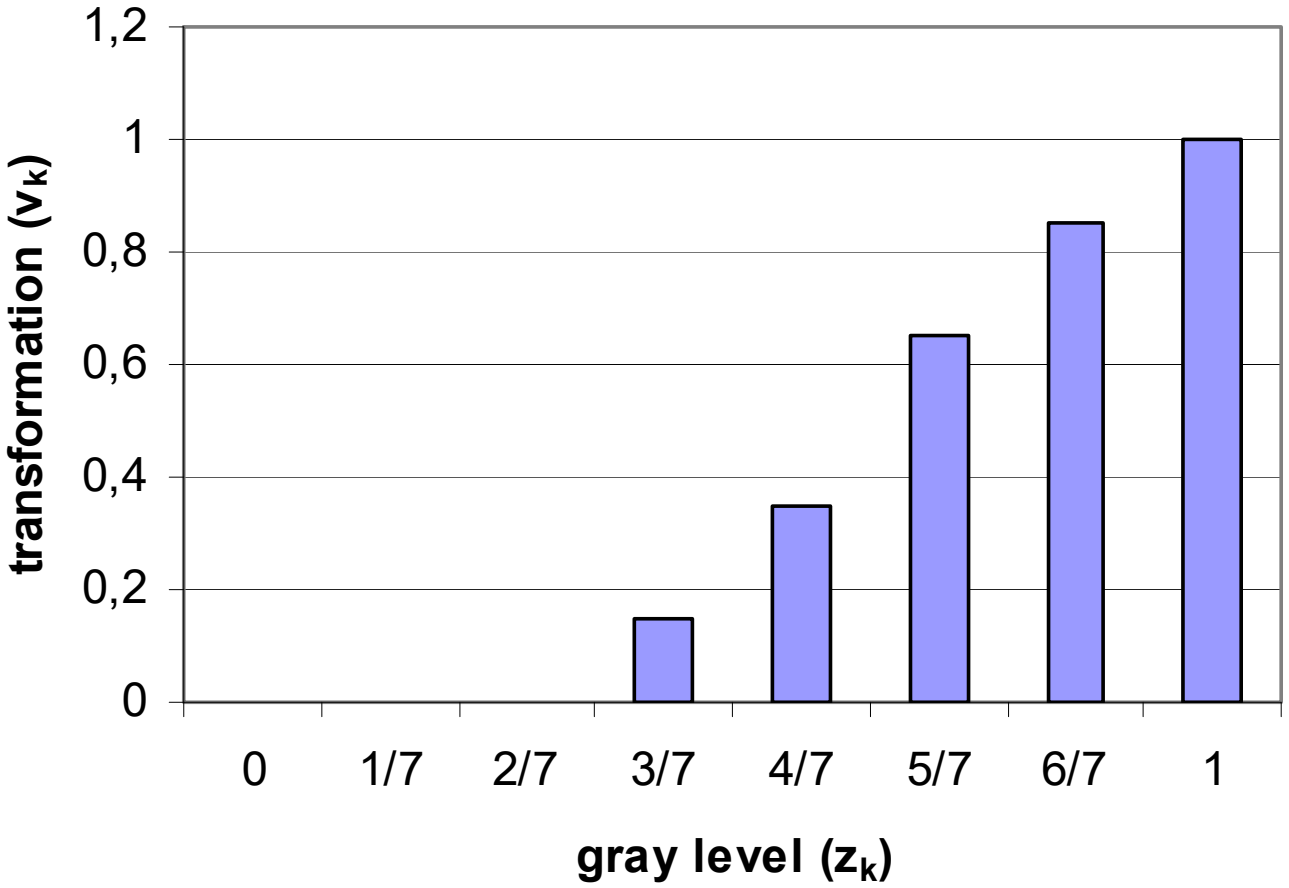
$$v_k = G(z_k) = \sum_{j=0}^{k} p_z(z_j)$$

- $v_0 = G(z_0) = 0{,}00$
- $v_1 = G(z_1) = 0{,}00$
- $v_2 = G(z_2) = 0{,}00$
- $v_3 = G(z_3) = 0{,}15$
- $v_4 = G(z_4) = 0{,}35$
- $v_5 = G(z_5) = 0{,}65$
- $v_6 = G(z_6) = 0{,}85$
- $v_7 = G(z_7) = 1{,}00$

Step 2

z_k	$p_z(z_k)$	V_k	V_k x 7	Normal(V_k)
$z_0=0$	0,00	0,00	0,00	$v_0=0$
$z_1=1/7$	0,00	0,00	0,00	$v_1=0$
$z_2=2/7$	0,00	0,00	0,00	$v_2=0$
$z_3=3/7$	0,15	0,15	$1,05 \cong 1$	$v_3=1/7$
$z_4=4/7$	0,20	0,35	$2,45 \cong 2$	$v_4=2/7$
$z_5=5/7$	0,30	0,65	$4,45 \cong 4$	$v_5=4/7$
$z_6=6/7$	0,20	0,85	$5.95 \cong 6$	$v_6=6/7$
$z_7=1$	0,15	1,00	7	$v_7=1$

Grafik fungsi transformasi



Step 3: inverse G s_k -> G(z_k

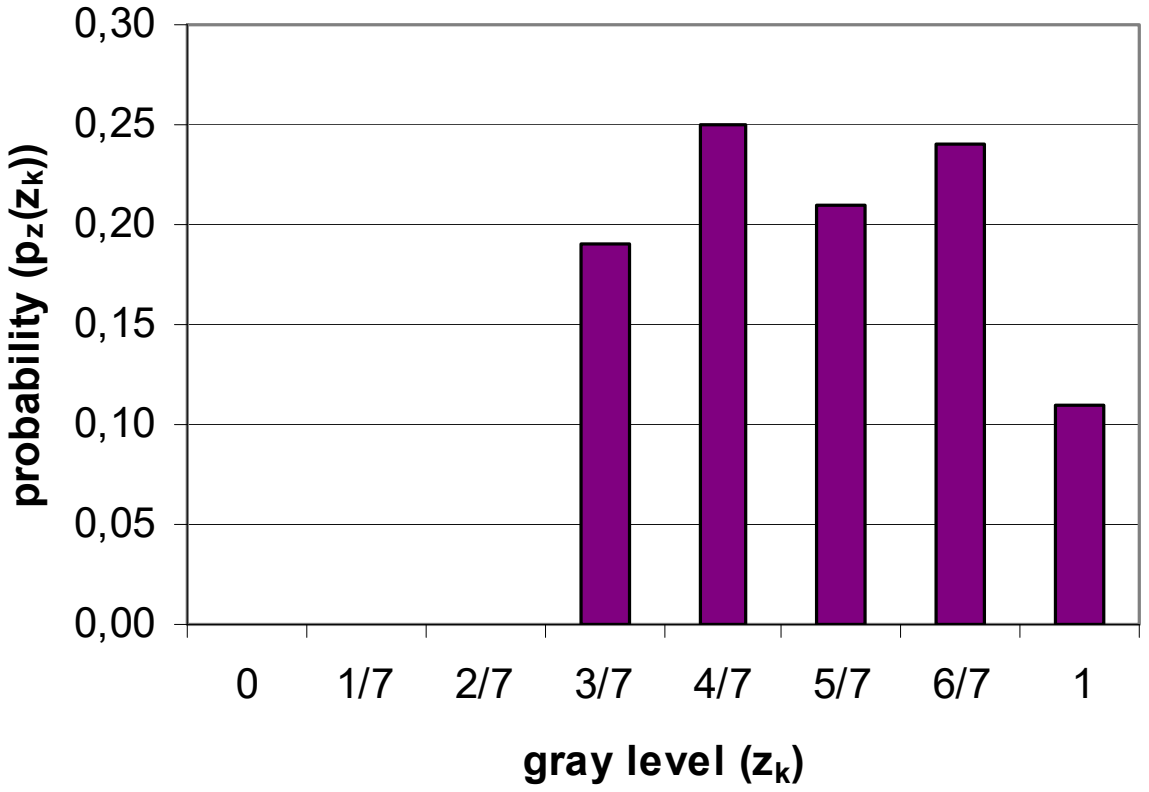
- $s_0 = 1/7 \approx 0.14 \rightarrow G(z_3) = 0.15;\ z_3 = 3/7$
- $s_1 = 3/7 \approx 0.43 \rightarrow G(z_4) = 0.35;\ z_4 = 4/7$
- $s_2 = 5/7 \approx 0.71 \rightarrow G(z_5) = 0.65;\ z_5 = 5/7$
- $s_3 = 6/7 \approx 0.86 \rightarrow G(z_6) = 0.85;\ z_6 = 6/7$
- $s_4 = 1 \rightarrow G(z_7) = 1.00;\ z_7 = 1$

Step 4: mapping r_k -> z_k

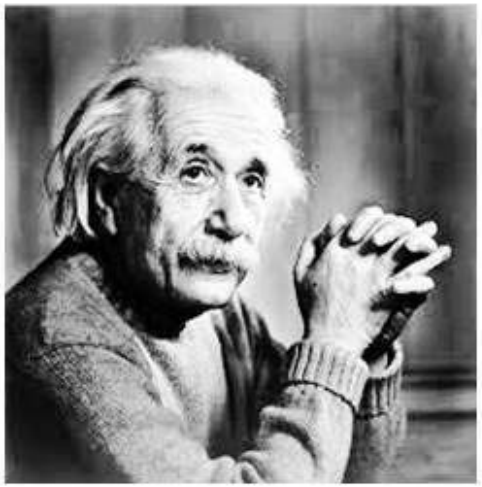
- $r_0 = 0 \rightarrow z_3 = 3/7$
- $r_1 = 1/7 \rightarrow z_4 = 4/7$
- $r_2 = 2/7 \rightarrow z_5 = 5/7$
- $r_3 = 3/7 \rightarrow z_6 = 6/7$
- $r_4 = 4/7 \rightarrow z_6 = 6/7$
- $r_5 = 5/7 \rightarrow z_7 = 1$
- $r_6 = 6/7 \rightarrow z_7 = 1$
- $r_7 = 1 \rightarrow z_7 = 1$

Output

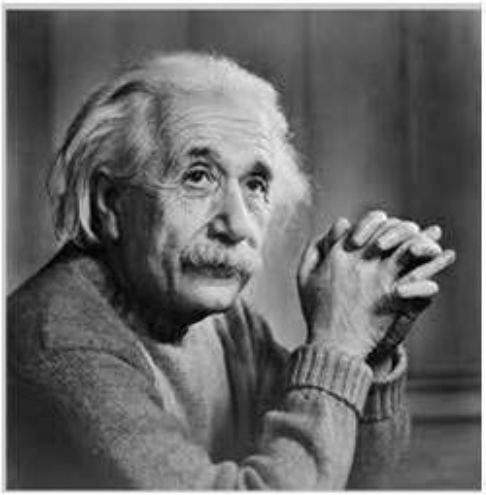
z_k	n_k	$p_z(z_k) = n_k/n$
$r_0 = 0$	0	0
$r_1 = 1/7$	0	0
$r_2 = 2/7$	0	0
$r_3 = 3/7$	790	0,19
$r_4 = 4/7$	1023	0,25
$r_5 = 5/7$	850	0,21
$r_6 = 6/7$	985	0,24
$r_7 = 1$	448	0,11



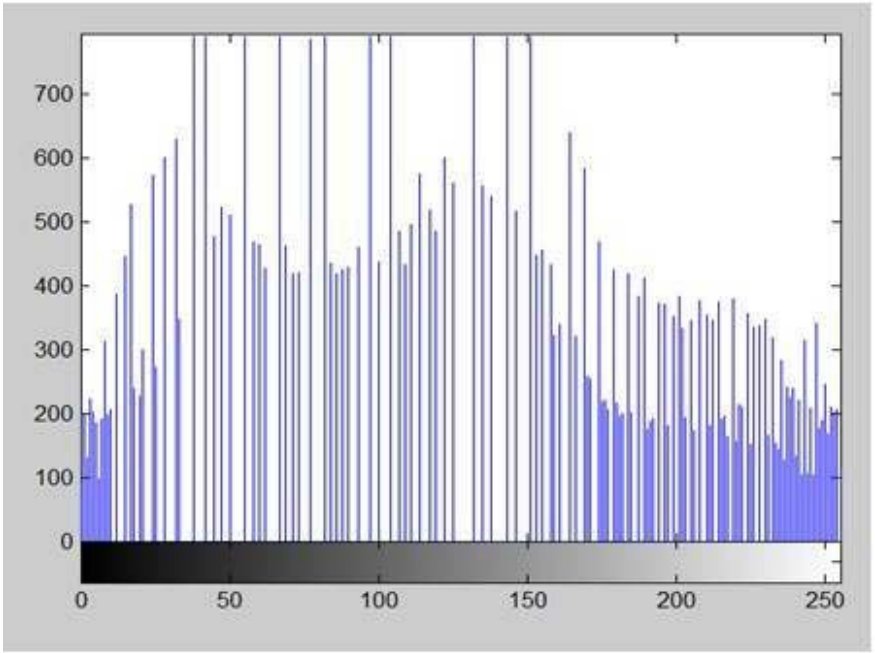
New Image



Old image



New Histogram



Old Histogram

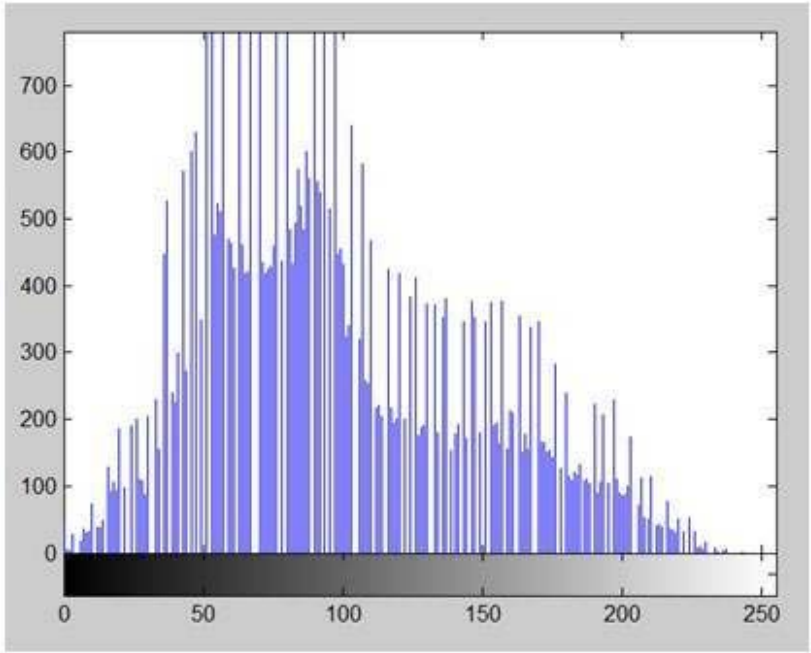
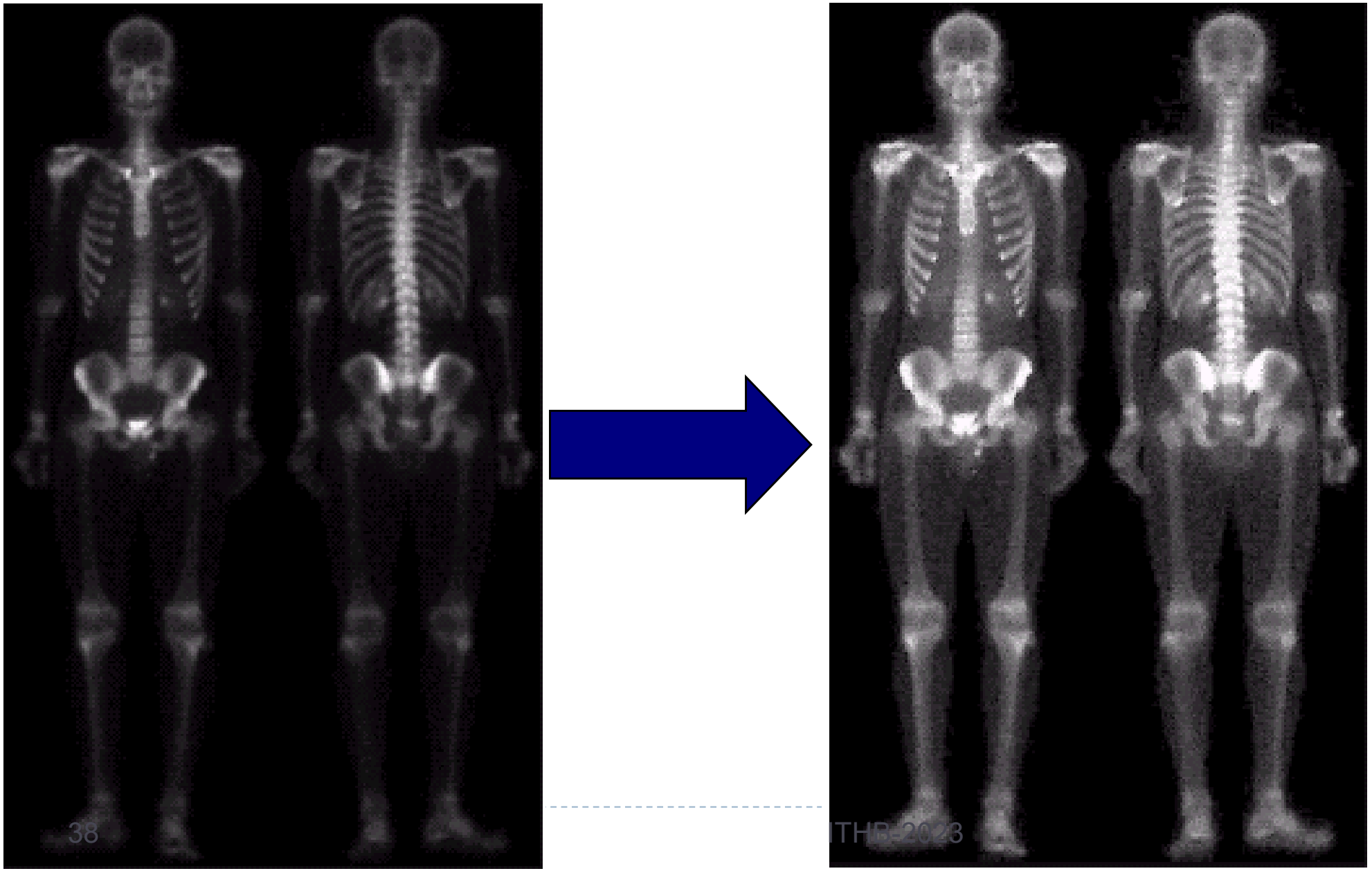
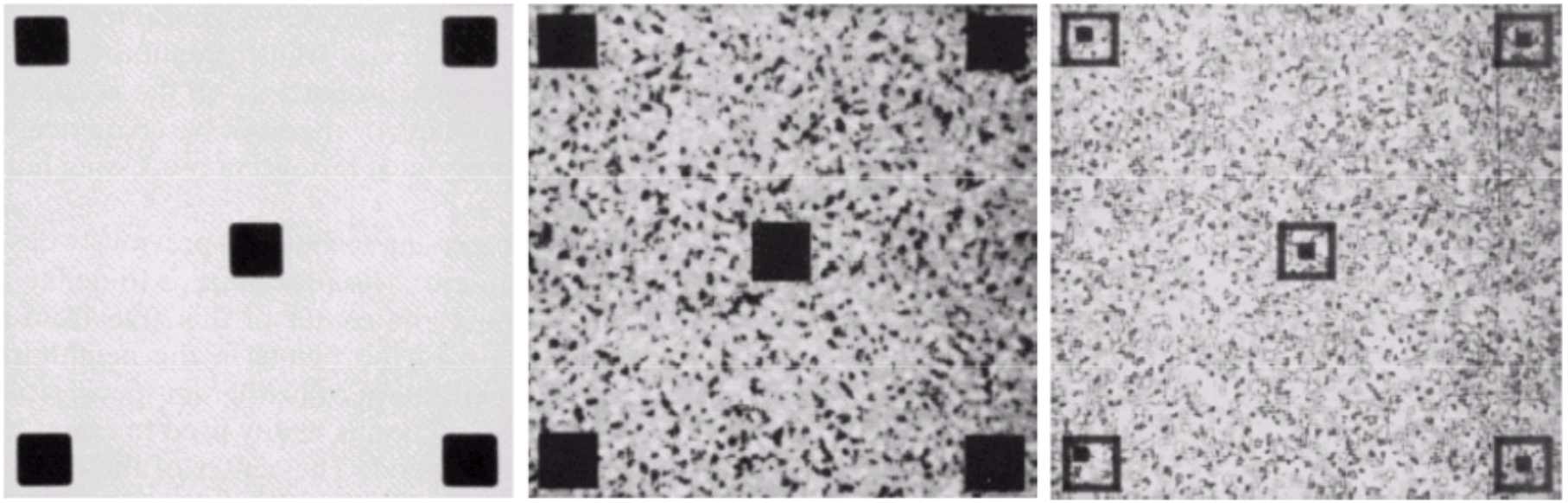


Image Enhancement Examples (cont…)



Local histogram Processing

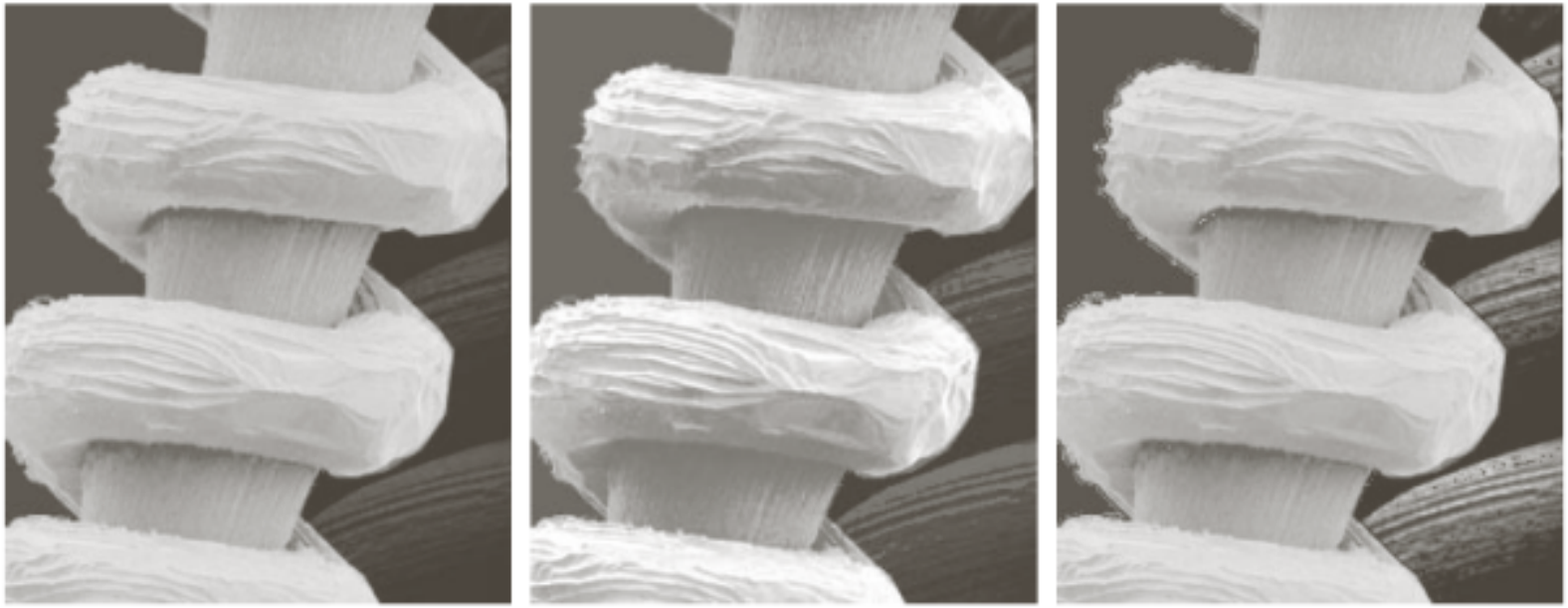
- Local histogram equalization



a b c

FIGURE 3.23 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization using a 7×7 neighborhood about each pixel.

Example



a | b | c

(a) original image

(b) equalized image

(c) local statistics enhanced image (3×3 neighborhood)

Histogram statistik

- Means and variance expressions from basic statistic :

$$m = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y)$$

$$\sigma^2 = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y) - m]^2$$

Summary

We have looked at:

- Different kinds of image enhancement
- Histograms
- Histogram equalisation
- Histogram specification

Next time we will start to look at spatial filtering and neighbourhood operations